UNIT 4: CLASSIFICATION

1. What is classification? Explain.
2. What is classification model? For what purposes classification model is used? Explain.
3. Brief following terms: Learning Algorithm, Training set, Test set.
4. Why confusion matrix is used? Give the structure of confusion matrix for a 2 class problem.
5. Which are the performance metrics used to compare different models? Explain.
6. Explain how decision tree works with the help of an example.
7. State and explain Hunt's algorithm for building decision trees with the help of an example.
8. Explain methods for extracting Attribute test condition.
9. Explain how Gini Index and Entropy can be used to select best split.
10. Give the structure of classification rule. Also explain how quality of a rule can be evaluated.
11. Explain how rule-based classifier works with the help of an example.
12. Explain properties of ruleset generated by a rule-based classifier. (OR what are Mutually exclusive and exhaustive rules?) Give example.
13. Explain the solution for mutual exclusiveness and exhaustiveness are not followed by rule set.
14. Explain two ways of implementing rule ordering with the help of an example.
15. Explain sequential covering algorithm used for rule extraction.
16. Compare eager and lazy classifiers.
17. Present the k-nearest neighbor classification algorithm. Explain same with the help of an example.
18. Explain the characteristics of Nearest Neighbor classifier.
19. State and explain Bayes Theorem with the help of an example.
20. Explain how the Bayes theorem is used for classification.
21. Explain different methods of evaluating the performance of a classifier.